FORTREAS FILM COATED TABLET 100MG

Sitagliptin (as Hydrochloride Monohydrate)
100mg

DESCRIPTION
Yellow, round, biconvex, film coated tablet with "100" on one side and scored on the other. Tablet can be split into half when necessary

THERAPEUTIC CLASS
Fortreas is an orally-active, potent, and highly selective inhibitor of the dipeptidyl peptidase 4 (DPP-4) enzyme for the treatment of type 2 diabetes. The DPP-4 inhibitors are a class of agents that act as incretin enhancers. By inhibiting the DPP-4 enzyme, sitagliptin increases the levels of two known active incretin hormones, glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP). The incretins are part of an endogenous system involved in the physiologic regulation of glucose homeostasis. When blood glucose concentrations are normal or elevated, GLP-1 and GIP increase insulin synthesis and release from pancreatic beta cells. GLP-1 also lowers glucagon secretion from pancreatic alpha cells, leading to reduced hepatic glucose production. This mechanism is unlike the mechanism seen with sulfonylureas; sulfonylureas cause insulin release even when glucose levels are low, which can lead to sulfonylurea-induced hypoglycaemia in patients with type 2 diabetes and in normal subjects. Sitagliptin is a potent and highly selective inhibitor of the enzyme DPP-4 and does not inhibit the closely-related enzymes DPP-8 or DPP-9 at therapeutic concentrations. Sitagliptin differs in chemical structure and pharmacological action from GLP-1 analogues, insulin, sulfonylureas or meglitinides, biguanides, peroxisome proliferator-activated receptor gamma (PPARγ) agonists, alpha-glucosidase inhibitors, and amylin analogues.

COMPOSITION
Active Ingredients
Each film coated of Fortreas Tablet contains 113.38 mg Sitagliptin Hydrochloride contains 100 mg of Sitagliptin.

MECHANISM OF ACTION
Fortreas is a member of a class of oral anti-hyperglycaemic agents called dipeptidyl peptidase 4 (DPP-4) inhibitors, which improve glycaemic control in patients with type 2 diabetes by enhancing the levels of active incretin hormones. Incretin hormones, including glucagon like peptide-1 (GLP1) and glucose dependent insulinotropic polypeptide (GIP) are released by the intestine throughout the day and levels are increased in response to a meal, The incretins are part of an endogenous system involved in the physiologic regulation of glucose homeostasis. When blood glucose concentrations are normal or elevated, GLP-1 and GIP increase insulin synthesis and release from pancreatic beta cells by intracellular signalling pathways involving cyclic AMP. Treatment with GLP-1 or with DPP-4 inhibitors in animal models of type 2 diabetes has been demonstrated to improve beta cell responsiveness to glucose and stimulate insulin biosynthesis and release. With higher insulin levels, tissue glucose uptake is enhanced. In addition, GLP-1 lowers glucagon secretion from pancreatic alpha cells. Decreased glucagon concentrations, along with higher insulin levels, lead to reduced hepatic glucose production, resulting in a decrease in blood glucose levels. The effects of GLP-1 and GIP are glucose dependent such that when blood glucose concentrations are low, stimulation of insulin release and suppression of glucagon secretion by GLP-1 are not observed. For both GLP-1 and GIP, stimulation of insulin release is enhanced as glucose rises above normal concentrations. Further, GLP-1 does not impair the normal glucagon response to hypoglycaemia. The activity of GLP -1 and GIP is limited by the DPP-4 enzyme, which rapidly hydrolyses the incretin hormones to produce inactive products. Sitagliptin prevents the hydrolysis of incretin hormones by DPP-4 thereby increasing plasma concentrations of the active forms of GLP-1 and GIP. By enhancing active incretins levels, sitagliptin increases insulin release and decreases glucagon levels in a glucose dependent manner. In patients with type 2 diabetes with hyperglycaemia, these changes in insulin and glucagon levels lead to lower haemoglobin A_{lc} (HBA_{lc}) and lower fasting and postprandial glucose concentrations. The glucose dependent mechanism of sitagliptin is distinct from the mechanism of sulfonylureas, which increase secretion even when glucose levels are low and can lead to hypoglycaemia in patients with type 2 diabetes and in normal subjects. Sitagliptin is a potent and highly selective inhibitor of the enzyme DPP-4 and does not inhibit the closely related enzymes DPP-8 or DPP-9 at therapeutic concentrations.

PHARMACODYNAMICS
General
In patients with type 2 diabetes, administration of single oral doses of Fortreas Tablet leads to inhibition of DPP-4 enzyme activity for a 24-hour period, resulting in a 2-to-3-fold increase in circulating levels of active GLP-1 and GIP, increased plasma levels of insulin and C-peptide, decreased glucagon concentrations, reduced fasting glucose and reduced glucose excursion following an oral glucose load or a meal.

Sitagliptin alone increased only active GLP-1 concentrations, reflecting inhibition of DPP-4, whereas metformin alone increased active and total GLP-1 concentrations to a similar extent.

PHARMACOKINETICS
Absorption
The absolute bioavailability of sitagliptin is approximately 87%. Since co-administration of a high fat meal with Fortreas Tablet had no effect on the pharmacokinetics, Fortreas may be administered with or without food.

Distribution
The mean volume of distribution of steady state following a single 100mg intravenous dose of sitagliptin to healthy subjects is approximately 198 litres. The fraction of sitagliptin reversibly bound to plasma is low 38%.

Metabolism
Sitagliptin is primarily eliminated unchanged in urine, and metabolism is a minor pathway. Approximately 79% of sitagliptin is excreted unchanged in the urine.

Following a [^{14}C] sitagliptin oral dose, approximately 16% of the radioactivity was excreted as metabolites of sitagliptin. Six metabolites were detected at trace levels and are not expected to contribute to the plasma DPP-4 inhibitory activity of sitagliptin. In vitro studies indicated that the primary enzyme responsible for the limited metabolism of sitagliptin was CYP3A4, with contribution from CYP2C8.

Elimination
Following administration of an oral [^{14}C] sitagliptin dose, approximately 100% of the administered radioactivity was eliminated in faeces (13%) or urine (87%) within one week of dosing. The apparent terminal $t_{½}$ following a 100mg oral dose of sitagliptin was approximately 12.4 hours and renal clearance was approximately 350mL/min.

Elimination of sitagliptin occurs primarily via renal excretion and involves active tubular secretion. Sitagliptin is a substrate for human organic anion transporter-3 (hOAT -3) which may be involved in the renal elimination of sitagliptin. The clinical relevance of hOAT-3 in sitagliptin transport has not been established. Sitagliptin is also a substrate of p-glycoprotein which may also be involved in mediating the renal elimination of sitagliptin. However, cyclosporine, a p-glycoprotein inhibitor, did not reduce clearance of sitagliptin.

Renal impairment
Sitagliptin was modestly removed by haemodialysis (13.5% over a 3- to 4-hour haemodialysis session starting 4 hours post dose). To achieve plasma concentrations of sitagliptin similar to those in patients with normal renal function, lower dosages are recommended in patients with moderate and severe renal insufficiency, as well as in ESRD patients requiring haemodialysis.

Hepatic insufficiently
No dosage adjustment for Fortreas Tablet is necessary for patients with mild or moderate hepatic insufficiency. There is no clinically experience in patients with severe hepatic insufficiency (Child- Pugh score >9). However, because sitagliptin is primarily renally eliminated, severe hepatic insufficiency is not expected to affect the pharmacokinetic of sitagliptin.

Elderly
No dosage adjustment is required based on age. Elderly subjects (65 to 85 years) had approximately 19% higher plasma concentrations of sitagliptin compared to younger subjects.

Paediatric
No studies with Fortreas have been performed in paediatric patients <10 years of age.

Gender
No dosage adjustment is necessary based on gender.

Race
No dosage adjustment is necessary based on race.

Body Mass Index (BMI)
No dosage adjustment is necessary based on BMI.

Type 2 diabetes
The pharmacokinetic of sitagliptin in patients with type 2 diabetes are generally similar to those in healthy subjects.

INDICATIONS
Monotherapy
Fortreas Tablet is indicated as an adjunct to diet and exercise to improve glycemic control in patients with type 2 diabetes mellitus.

Combination with metformin
Fortreas Tablet is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with metformin as initial therapy or when the single agent alone, with diet and exercise does not provide adequate glycemic control. Initial combination therapy or maintenance of combination therapy may not be appropriate for all patients. These management options are left to the discretion of the health care provider.

Combination with a sulphonylurea
Fortreas Tablet is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with sulphonylurea when treatment with maximal tolerated dose of sulphonylurea alone, with diet and exercise, does not provide adequate glycemic control and when metformin is inappropriate due to contraindications or intolerance.

Combination with metformin and sulphonylurea
Fortreas Tablet is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with metformin and a sulphonylurea when dual therapy with these two agents and with diet and exercise does not provide adequate glycemic control.

Combination with peroxisome proliferator-activated receptor gamma (PPARγ) agonist
Fortreas Tablet is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with a PPARγ agonist (i.e., thiazolidinediones) when diet and exercise, plus the single agent do not provide adequate glycemic control.

Combination with metformin and a PPARγ agonist
Fortreas Tablet is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with metformin and PPARγ agonist (i.e thizolidinediones) when dual therapy with these agents, with diet and exercise does not provide adequate glycemic control.

Combination with insulin
Fortreas is indicated in patients with type 2 diabetes mellitus as an adjunct to diet and exercise to improve glycemic control in combination with insulin (with or without metformin).

DOSAGE AND ADMINISTRATION
The recommended dose for Fortreas Tablet is 100 mg once daily as monotherapy or as combination therapy with metformin, a sulfonylurea, insulin (with or without metformin), PPARγ agonist (i.e thiazolidinediones) metformin plus a sulfonylurea or metformin plus a PPARγ agonist Fortreas Tablet can be taken with or without food.

When Fortreas Tablet is used in combination with a sulfonylurea or with insulin, a lower dose of sulfonylurea or insulin may be considered to reduce the risk of sulfonylurea or insulin induced hypoglycemia (See PRECAUTIONS, hypoglycemia in combination with a sulfonylurea or with insulin)

Patients with renal impairment
Because there is a dosage adjustment based upon renal function, assessment of renal function is recommended prior to initiation of Fortreas and periodically thereafter.

For patients with mild renal impairment (estimated glomerular filtration rate [eGFR] ≥60 mL/min/1.73 m2 to < 90 mL/min/1.73 m2), no dosage adjustment for Fortreas is required.

For patients with moderate renal impairment (eGFR ≥ 45 mL/min.1.73 m2 to < 60 mL/min.1.73 m2), no dosage adjustment for Fortreas is required.

For patients with moderate renal impairment (eGFR ≥ 30 mL/min/1.73 m2 to 45 mL/min/1.73 m2), the dose of Fortreas is 50 mg once daily.

For patients with severe renal impairment (eGFR ≥ 15 mL/min/1.73 m2 to < 30 mL/min/1.73 m2) or with end-stage renal disease (ESRD) (eGFR < 15 mL/min/1.73m2), including those requiring hemodialysis or peritoneal dialysis, the dose of Fortreas is 25 mg once daily. Fortreas may be administered without regard to the timing of dialysis.

Pediatric population
Fortreas should not be used in children and adolescents 10 to 17 years of age because of insufficient efficacy. Fortreas has not been studied in pediatric patients under 10 years of age.

CONTRAINDICATIONS
Fortreas is contraindicated in patients who are hypersensitive to any components of this product. (See PRECAUTIONS, Hypersensitivity Reactions and SIDE EFFECTS)

PRECAUTIONS
General: Fortreas should not be used in patients with type 1 diabetes or for the treatment of diabetic ketoacidosis.

Pancreatitis: There have been reports of acute pancreatitis, including fatal and non-fatal haemorrhagic or necrotizing pancreatitis (see SIDE EFFECTS) in patients taking sitagliptin. Patients should be informed of the characteristic symptom of acute pancreatitis; persistent, severe abdominal pain. Resolution of pancreatitis has been observed after discontinuation of sitagliptin. If pancreatitis is suspected Fortreas and other potentially suspect medicinal products should be discontinued.

Use in Patients with Renal Insufficiency: Fortreas is renally excreted. To achieve plasma concentrations of Fortreas Tablet similar to those in patients with normal renal function; lower dosages are recommended in patients with moderate and severe renal insufficiency, as well as in ESRD patients requiring haemodialysis or peritoneal dialysis. (See DOSAGE AND ADMINISTRATION, Patients with renal insufficiency)

Hypoglycaemia in Combination with Sulfonylurea or with Insulin: As is typical with other antihyperglycaemic agents, hypoglycaemia is expected to be observed when Fortreas was used in combination with insulin or a sulfonylurea (see SIDE EFFECTS). Therefore, to reduce the risk of sulfonylurea- or insulin- induced hypoglycaemia, a lower dose of sulfonylurea or insulin may be considered (see DOSAGE AND ADMINISTRATION)

Hypersensitivity Reactions: The reactions include anaphylaxis, angioedema and exfoliative skin conditions including Stevens- Johnson syndrome. If a hypersensitivity reaction is suspected, discontinue Fortreas, assess for other potential causes for the event, and institute alternative treatment for diabetes (See CONTRAINDICATIONS and SIDE EFFECTS)

Severe and disabling arthralgia: Consider DPP-4 inhibitors as a possible cause of severe joint pain and discontinue drug if appropriate.

Bullous Pemphigoid: Tell patients to report development of blisters or erosions while receiving Fortreas. If bullous pemphigoid is suspected, Fortreas should be discontinued and referral to a dermatologist should be considered for diagnosis and appropriate treatment.

PREGNANCY
There are no adequate and well controlled studies in pregnant women, therefore the safety of Fortreas in pregnant women is not known. Fortreas, like other oral antihyperglycaemic agents is not recommended for use in pregnancy.

NURSING MOTHER
Sitagliptin is secreted in the milk of lactating rats. It is not known whether sitagliptin is secreted in human milk. Therefore, Fortreas should not be used by a woman who is nursing.

PEDIATRIC USE
Fortreas has not been studied in paediatric patients under 10 years of age.

USE IN THE ELDERLY
No dosage adjustment is required based on age. Elderly patients are more likely to have renal impairment; as with other patients, dosage adjustment may be required in the presence of significant renal impairment (see DOSAGE AND ADMINISTRATION, patient with renal impairment)

DRUG INTERACTION
Patients receiving digoxin should be monitored appropriately. No dosage adjustment of digoxin or Fortreas Tablet is recommended.

No dosage adjustment for Fortreas is recommended when co-administered with cyclosporine or other p-glycoprotein inhibitors (e.g ketoconazole).

SIDE EFFECTS
Fortreas was generally well tolerated, however below are some side effects which may encountered when taking Fortreas-:

Side Effects Reaction	Frequency of Side Effects
Blood and lymphatic system disorders	
thrombocytopenia	Rare
Immune system disorders	
hypersensitivity reactions including anaphylactic responses	Frequency not known
Metabolism and nutrition disorders	
hypoglycaemia	Common
Nervous system disorders	
headache	Common
dizziness	Uncommon
Respiratory, thoracic and mediastinal disorders	
interstitial lung disease	Frequency not known
Gastrointestinal disorders	
constipation	Uncommon
vomiting	Frequency not known
acute pancreatitis	Frequency not known
fatal and non-fatal haemorrhagic and necrotizing pancreatitis	Frequency not known
Skin and subcutaneous tissue disorders	
pruritus	Uncommon
angioedema	Frequency not known

rash	Frequency not known
urticaria	Frequency not known
cutaneous vasculitis	Frequency not known
exfoliative skin conditions including Stevens-Johnson syndrome	Frequency not known
bullous pemphigoid	Frequency not known
Musculoskeletal and connective tissue disorders	
arthralgia	Frequency not known
myalgia	Frequency not known
back pain	Frequency not known
arthropathy	Frequency not known
Renal and urinary disorders	
impaired renal function	Frequency not known
acute renal failure	Frequency not known

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Fortreas has no or negligible influence on the ability to drive and use machines. However, when driving or using machines, it should be taken into account that dizziness and somnolence have been reported.

In addition, patients should be alerted to the risk of hypoglycaemia when Fortreas is used in combination with a sulphonylurea or with insulin.

OVERDOSE

In the event of an overdose, it is reasonable to employ the usual supportive measures e.g remove unabsorbed material from the gastrointestinal tract, employ clinical monitoring (including obtaining an electrocardiogram and institute supportive therapy if required. Sitagliptin is modestly dialyzable. Prolonged hemodialysis may be considered if clinically appropriate. It is not known if sitagliptin is dialyzable by peritoneal dialysis.

ROUTE OF ADMINISTRATION

Oral.

SHELF LIFE

Please refer to outer package.

PACK SIZES

to 3x10's Alu-PVC/PE/PVDC Blister Pack.

STORAGE CONDITION

Store in a dry place 30°C. Protect from light.
Keep out of reach of children.
Jauhkan daripada kanak-kanak.

PRODUCT REGISTRATION HOLDER

Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2, 4, 6, 8, 10, Jalan P/7, Section 13,
Bangi Industrial Estate, 43650,
Bandar Baru Bangi, Selangor, Malaysia.

MANUFACTURER

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DATE OF REVISION

August 2022

Remarks:
Leaflet is preliminary drafted in Calibri with Font size of 11 which subject to change to any equivalent font and readable font size.